ASWINRAJA R

240701824

Experiment No. 1: Form Creation

1. Aim

To create a form in Zoho Creator using different field types, apply validation to the fields and use a lookup field to connect related information.

2. Introduction

Zoho Creator is a low-code application development platform used to create applications with forms, reports and workflows. A form is used to collect information from users and store it as records in the application. Different field types can be added according to the data that has to be collected.

In this experiment, a simple **Add New Lead** form is created. The form contains text, email, dropdown, date, currency, lookup and multi-line fields. Validation is used for important fields so that incorrect or incomplete data is not submitted.

3. Form Details

Form Name	Add New Lead
Application	My first Creator app
Purpose	To store details of potential customers (leads)
Platform	Zoho Creator

4. Fields Used

The following fields are used in the Add New Lead form shown in the Zoho Creator interface.

S.No	Field Label	Field Type	Validation / Lookup	Description
1	Prospect Name	Single Line	Mandatory	Name of the lead
2	Contact Info	Email	Valid email format	Email ID of the lead
3	Lead Source	Dropdown	-	Source of the lead
4	Opportunity Stage	Dropdown	-	Current stage of the opportunity
5	Forecasted Close Date	Date	-	Expected closing date
6	Potential Revenue	Currency / Selection	Value should be valid	Estimated revenue
7	Sales Rep	Lookup	Lookup to Sales Rep form	Select sales representative
8	Notes	Multi Line	-	Additional notes

5. Validation

Validation is applied to the fields where the entered value has to follow a particular condition. Prospect Name is kept mandatory, Contact Info uses email-format validation, and Potential Revenue is checked for a valid value. These checks help prevent incorrect data from being stored.

6. Lookup Field

Sales Rep is created as a Lookup field. It gets the available sales representatives from a related Sales Rep form. Therefore, the user can select an existing sales representative instead of entering the name manually.

7. Procedure

1. Login to Zoho Creator and open the application.

2. Go to the Design section of the application.

3. Click Add New and select Form.

4. Choose Blank Form / create the form from scratch.

5. Enter the form name as Add New Lead and create the form.

6. Add a Single Line field and name it Prospect Name. Set it as mandatory.

7. Add an Email field and name it Contact Info.

8. Add a Dropdown field for Lead Source and enter suitable options such as Website, Referral and Social Media.

9. Add another Dropdown field for Opportunity Stage and enter the required stages.

10. Add a Date field and name it Forecasted Close Date.

11. Add a Currency / suitable numeric field for Potential Revenue.

12. Create a Sales Rep form with the required sales representative records.

13. Add a Lookup field named Sales Rep and connect it to the Sales Rep form.

14. Add a Multi Line field named Notes.

15. Open the field properties and configure mandatory and validation settings.

16. Arrange the fields in the form and save the form.

17. Open the live form and enter sample lead information.

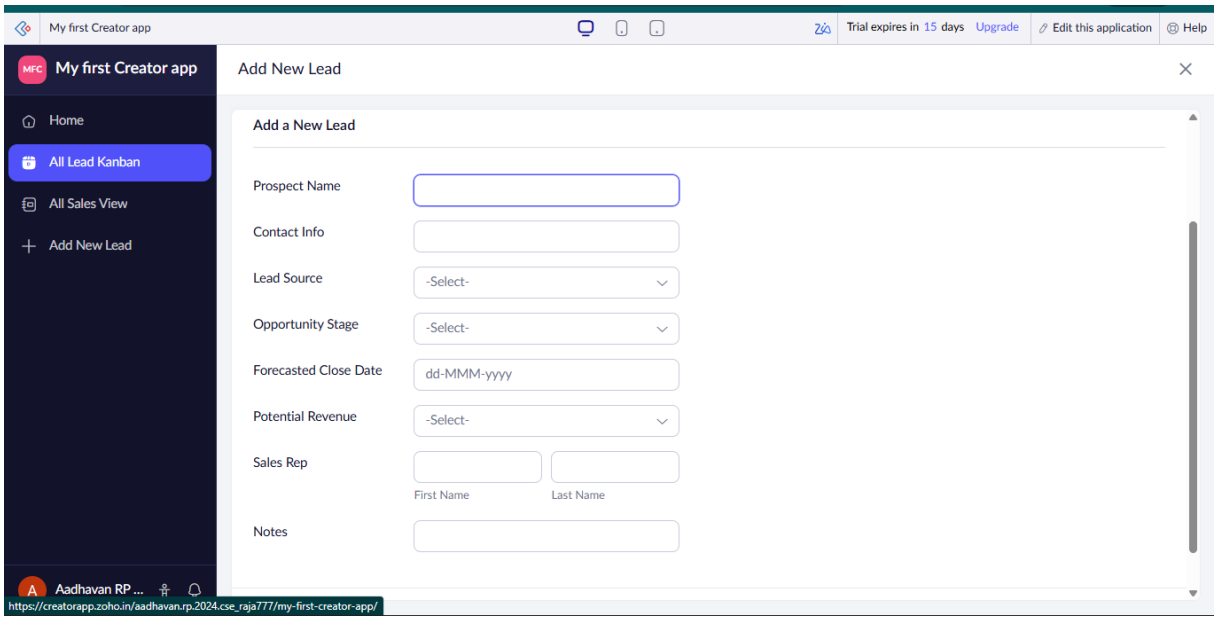
18. Check the dropdown and lookup values and click Submit.

8. Sample Data Used

Field	Sample Value
Prospect Name	Aadhavan Technologies
Contact Info	aadhavan@example.com
Lead Source	Website
Opportunity Stage	Qualification
Forecasted Close Date	30-Sep-2026
Potential Revenue	■ 50,000
Sales Rep	ASWINRAJA R
Notes	Interested in product demonstration

9. Output

The output below is based directly on the Zoho Creator page supplied as the reference. It shows the Add New Lead form with the left-side application menu, form heading, input boxes, dropdowns, date field, sales representative fields and notes field.



Observation: The Add New Lead form is displayed in the Zoho Creator application. The user can enter the lead information and select values from the available dropdown and lookup controls.

10. Validation Test

Test	Action	Expected Result
1	Leave Prospect Name empty	Mandatory validation is shown
2	Enter an invalid email in Contact Info	Email validation is shown
3	Select a value from Lead Source	Selected value is displayed in the field
4	Select Sales Rep	Existing Sales Rep record can be selected through lookup
5	Enter valid lead details and submit	Lead record is saved successfully

11. Result

Thus, the Add New Lead form was created successfully in Zoho Creator. Different field types were added, validation was applied to the required fields, and the Sales Rep field was connected using a lookup. The form was tested with sample data and the record was submitted successfully.

ASWINRAJA R

240701824

Experiment No. 2: Comparative Study of Low-Code Development Versus Traditional Application Development

1. Aim

To study and compare the Traditional Application Development approach with the Low-Code Development approach based on their development process, speed, cost, flexibility and ease of use.

2. Introduction to Traditional Application Development

Traditional application development is the conventional method of building software by writing source code using programming languages such as Java, Python, C# or JavaScript. Developers manually design, code, test, deploy and maintain the application.

This approach provides complete control over the application's functionality, performance and customization. However, it requires skilled developers, longer development time and higher development costs.

Features

• Manual coding is required.
• High flexibility and customization.
• Suitable for complex applications.
• Requires experienced developers.
• Longer testing and deployment cycles.

3. Introduction to Low-Code Development

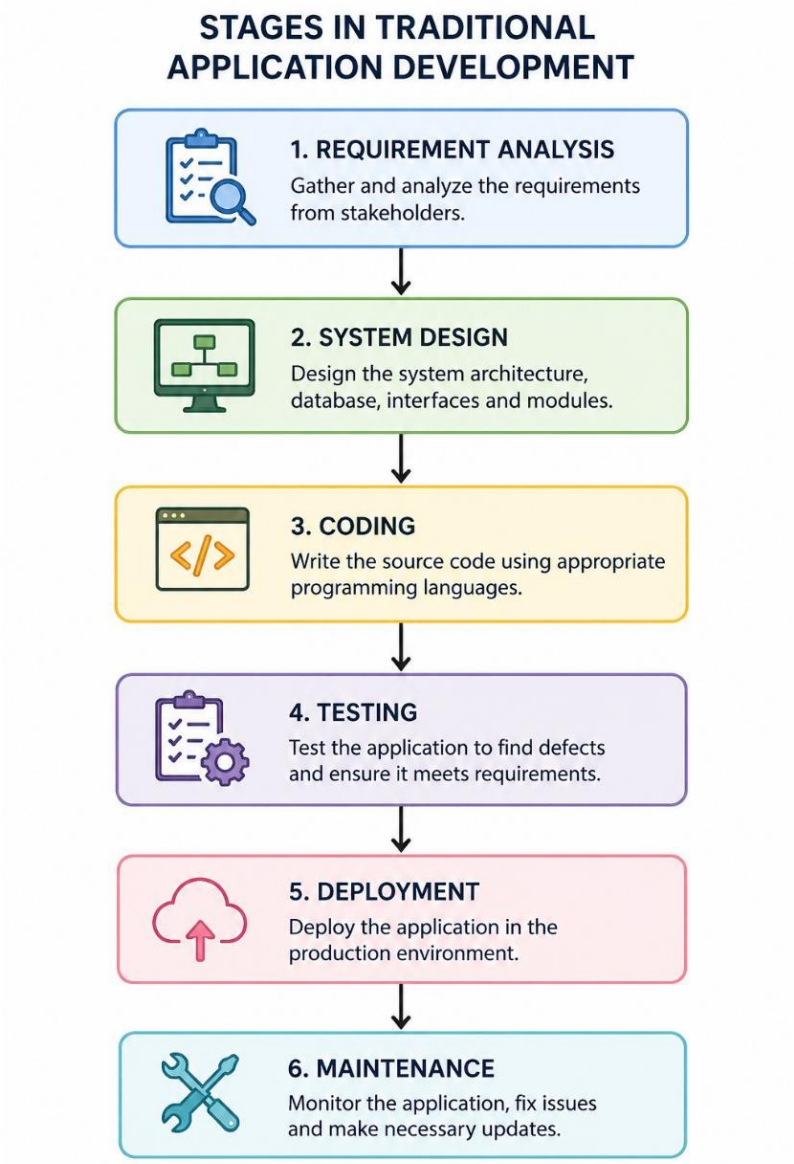
Low-code development is a modern software development approach that enables applications to be built using graphical user interfaces, drag-and-drop components and minimal hand-written code.

It allows professional developers and non-technical users to create applications quickly, reducing development time and effort.

Features

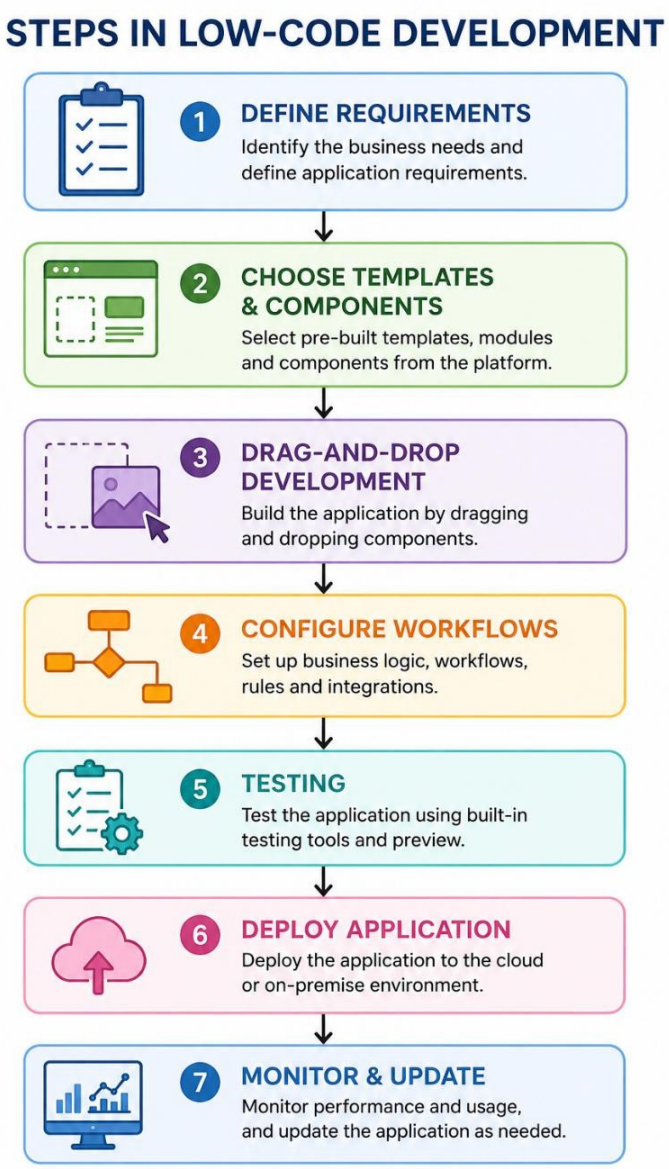
• Drag-and-drop interface.
• Minimal coding required.
• Faster application development.
• Easy integration with existing systems.
• Suitable for rapid application development and business applications.

4. Diagram for Stages in Traditional Application Development



The stages shown are requirement analysis, system design, coding, testing, deployment and maintenance.

5. Diagram for Steps in Low-Code Development



The low-code process includes defining requirements, choosing templates and components, drag-and-drop development, configuring workflows, testing, deployment and monitoring/updating.

6. Comparative Analysis of Traditional and Low-Code Development

Parameter	Traditional Development	Low-Code Development
Coding	Extensive manual coding	Minimal coding
Development Speed	Slow	Very Fast
Development Cost	High	Low
Required Skills	Professional programmers	Developers and non-developers
Customization	Very High	Moderate to High
Learning Curve	Steep	Easy
Maintenance	Time-consuming	Easier
Testing	Mostly manual	Automated tools available
Deployment	Takes more time	Quick deployment
Best Suitable For	Large and complex applications	Business applications and rapid development

7. Main Differences

Development: Traditional development depends mainly on manual programming, whereas low-code development uses visual components and less handwritten code.

Time and Cost: Traditional development generally requires more development and testing effort. Low-code platforms are designed to reduce development time and effort.

Skills: Traditional development normally requires programming knowledge. Low-code platforms can also support users with less programming experience.

Customization: Traditional development gives extensive control over application behaviour. Low-code platforms provide configurable components and can also support additional logic where required.

8. Advantages and Limitations

Traditional Development	Low-Code Development
High control over application architecture	Faster application development
Suitable for highly customized systems	Drag-and-drop development
Suitable for complex applications	Less manual coding
Requires skilled developers	Can be used by developers and non-developers
Longer development and maintenance effort	Simplifies many development tasks

9. Result

The comparative study was completed. Traditional Application Development provides extensive control and customization but generally involves more manual coding, development effort and skilled programming. Low-Code Development uses visual components and minimal coding to simplify and speed up application development.

The suitable approach depends on the application requirements, complexity, available skills, budget and development time.

EXPERIMENT NO. 3

Application Building using Zia Assistant

Name: ASWINRAJA R **Register No:** 240701824

AIM

To create a functional College Event Registration System in Zoho Creator using Zia Assistant by giving the application requirements in natural language.

INTRODUCTION

Zoho Creator is a low-code application development platform. Zia Assistant helps users create an application from a natural-language description. By providing the required forms, fields, reports, dashboard elements and workflow requirements, Zia can generate the initial application structure automatically.

APPLICATION CREATED

College Event Registration System

The application is used to manage students, college events and registrations. It allows the user to add student details, create events, register students for events, maintain registration status and monitor the overall activity through a dashboard.

PROMPT GIVEN TO ZIA

Create a College Event Registration System. Add Students, Events and Registrations modules. Students should contain Student Name, Register Number, Department, Email, Phone Number and Year. Events should contain Event Name, Event Type, Event Date, Venue, Coordinator, Maximum Participants and Description. Registrations should connect Students and Events using lookup fields and include Registration Date, Registration Status and Remarks. Create reports for students, events and registrations. Create a dashboard showing Confirmed, Cancelled and Pending registrations, year-wise information, event type statistics and quick links for adding records.

MAIN COMPONENTS

Component	Purpose
Students	Stores student information and academic year.
Events	Stores event details, type, venue and participant limit.
Registrations	Links students with events and stores registration status.
Reports	Displays stored student, event and registration records.
Dashboard	Shows registration statistics, charts and quick links.

PROCEDURE

1. Open Zoho Creator and choose the option to create a new application.
2. Select **Create using Zia**.
3. In the Zia App Builder screen, keep the mode as **Build**.
4. Enter the College Event Registration System requirements in the prompt box.
5. Submit the prompt and allow Zia to analyze the requirement.
6. Review the forms, fields, lookup relationships and reports suggested by Zia.
7. Create the application from the generated structure.
8. Open the generated Students module and enter sample student records.
9. Open the Events module and enter event details.
10. Open the Registrations module and select a Student and Event using the generated lookup fields.
11. Enter the Registration Date, Registration Status and Remarks.
12. Submit the registration and verify that the record appears in the registration report.
13. Open the Dashboard and verify Confirmed, Cancelled, Pending and total registration values.
14. Verify the year-wise and event-type visualizations and use the quick links for adding records.

FORM DETAILS

Form	Important Fields
Students	Student Name, Register Number, Department, Email, Phone Number, Year
Events	Event Name, Event Type, Event Date, Venue, Coordinator, Maximum Participants, Description
Registrations	Student, Event, Registration Date, Registration Status, Remarks

LOOKUP RELATIONSHIP

The Registrations form uses lookup fields for Student and Event. This connects each registration record with an existing student and an existing event and avoids repeatedly entering the same student or event details.

REGISTRATION STATUS

The Registration Status field is used to identify whether a registration is Confirmed, Pending or Cancelled. These values are also summarized in the dashboard.

APPLICATION STRUCTURE AND TEST DATA

DATA MODEL

Module	Sample Fields	Relation
Students	Student Name, Register Number, Department, Year	Referenced by Registrations
Events	Event Name, Event Type, Event Date, Venue	Referenced by Registrations
Registrations	Student, Event, Registration Date, Status	Connects Students and Events

SAMPLE DATA USED

Record	Sample Value
Student	STU001 - ASWINRAJA R
Department	CSE
Year	3rd Year
Event	Technical Event / Hackathon
Registration Status	Confirmed
Registration Date	Entered through the generated registration form

DASHBOARD ELEMENTS GENERATED

The generated dashboard contains summary cards for Confirmed, Cancelled, Pending and Registration Status. It also contains a Year section, a chart showing Event Type by Maximum Participants, a Distribution by Student chart and quick links for adding new registrations, students and events.

WORKING OF THE APPLICATION

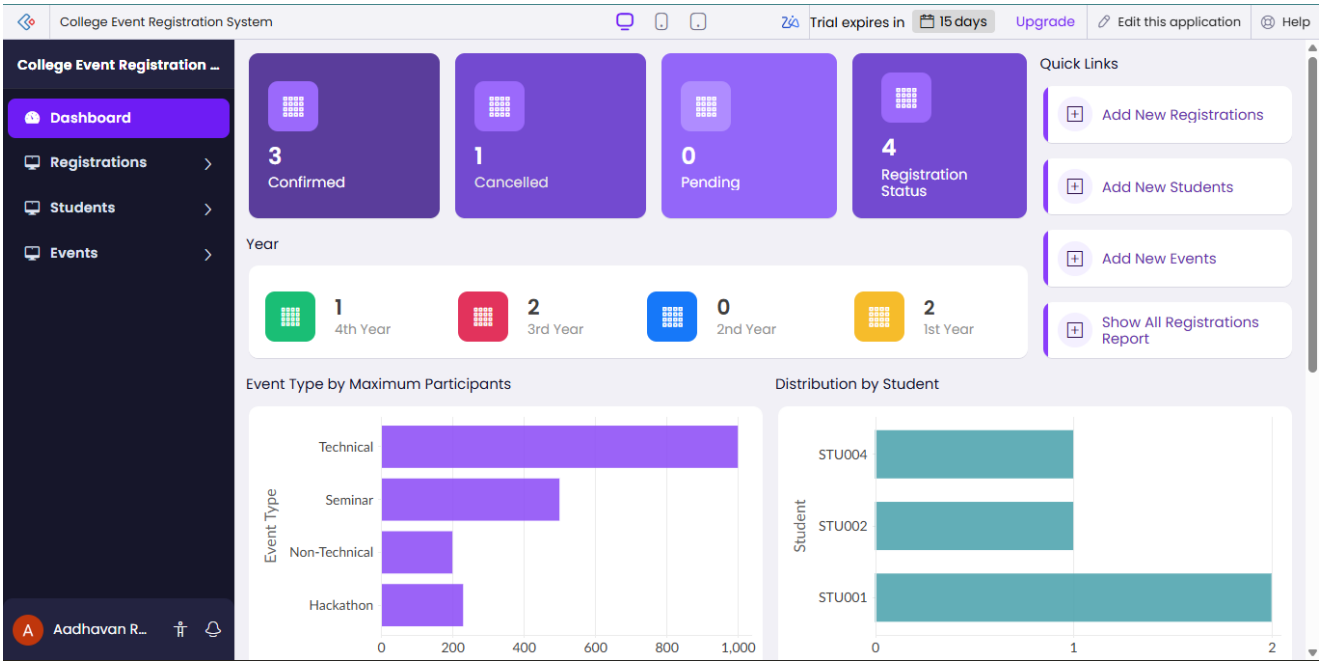
A user first creates student and event records. While adding a registration, the Student and Event fields display the existing records through lookup controls. After the registration is submitted, it is available in the registration report and its status contributes to the dashboard statistics. This provides a single interface for entering and monitoring college event registration data.

VALIDATION / TESTING

Test	Expected Result	Status
Open Registrations form	Student and Event lookup fields are displayed.	Pass
Submit registration	Registration record is saved.	Pass
Open Dashboard	Registration summary cards are visible.	Pass
Check charts	Year and event-related visualizations are displayed.	Pass
Use quick links	Links open the corresponding add/report screens.	Pass

OUTPUT - DASHBOARD

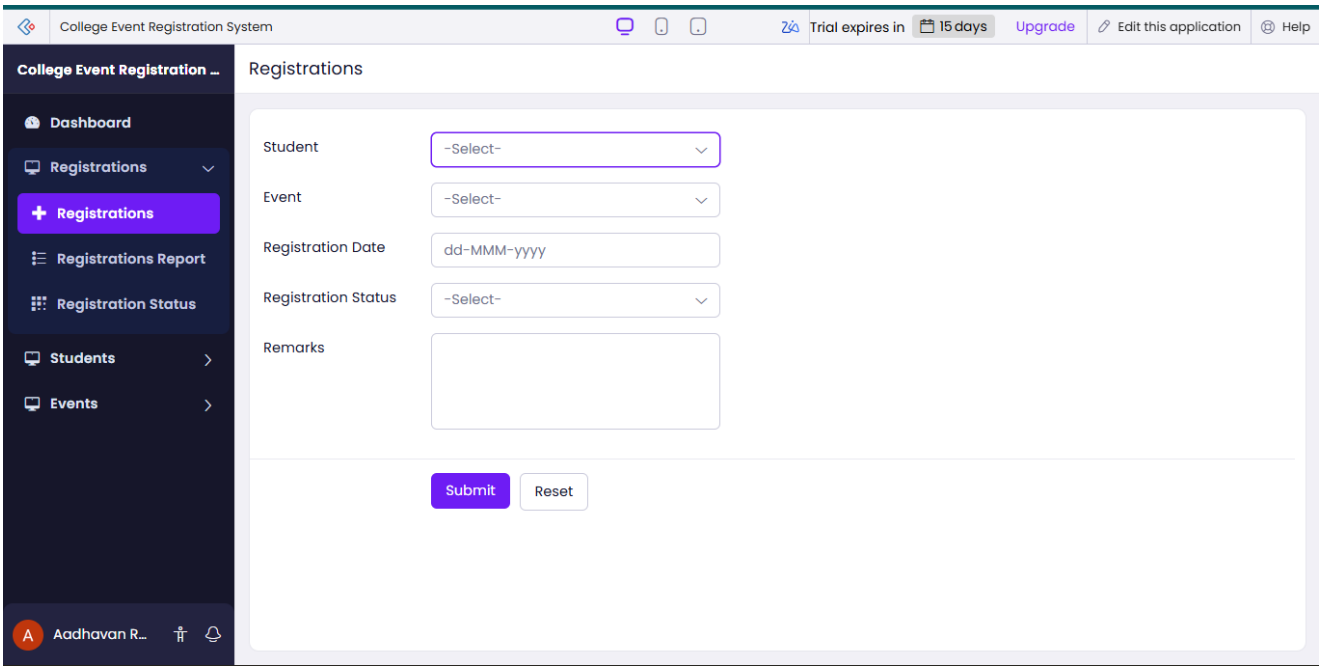
The following is the actual output of the College Event Registration System generated in Zoho Creator using Zia.



Output Observation: The dashboard displays Confirmed, Cancelled and Pending registration counts, a total Registration Status card, year-wise student information, Event Type by Maximum Participants, Distribution by Student and quick links for adding new registrations, students and events.

OUTPUT - REGISTRATION FORM

The generated Registrations form contains lookup and data-entry fields required for event registration.



Output Observation: The registration form contains Student, Event, Registration Date, Registration Status and Remarks fields, along with Submit and Reset buttons. The Student and Event fields are shown as selection controls so that existing records can be linked.

RESULT

Thus, the College Event Registration System was successfully created using Zoho Creator Zia Assistant. The generated application contains Students, Events and Registrations modules, reports and a dashboard. The registration form and dashboard were tested successfully with sample data.

EXPERIMENT NO. 4

Reports in Zoho Creator

Name: ASWINRAJA R **Register No:** 240701824

AIM

To create and customize reports in Zoho Creator for displaying, filtering, grouping, sorting and analysing the records collected through application forms.

INTRODUCTION

A report in Zoho Creator is used to present the records collected through a form in a meaningful way. Reports help users view, search, filter, group and analyse application data. Different report layouts can be selected depending on the type of information that needs to be displayed.

APPLICATION USED

College Event Registration System

The reports for this experiment are based on the Students, Events and Registrations modules of the College Event Registration System.

REPORTS CREATED

Report	Based On	Purpose
Students Report	Students	Display student records.
Department Report	Students	Group students according to department.
Registrations Report	Registrations	Display event registration records.
Registration Status Report	Registrations	Summarize registration status.
Events Report	Events	Display event information and dates.

REPORT TYPES AND PROCEDURE

REPORT TYPES

Report Type	Description	Example
List Report	Displays records in rows, columns or cards.	Students Report
Calendar Report	Displays records according to date.	Events Report
Kanban Report	Groups records according to a selected field.	Registration Status
Pivot Table	Summarizes records using rows and columns.	Department / Year
Pivot Chart	Represents summarized information graphically.	Registration statistics
Spreadsheet Report	Displays data in a spreadsheet-like layout.	Student analysis

PROCEDURE

1. Open the College Event Registration System in Zoho Creator.
2. Open the Design page of the application.
3. Select the required form, such as Students or Registrations.
4. Create a new report and select the required report type.
5. Give the report a suitable name.
6. Select the fields that must be displayed in the report.
7. Configure filters, grouping and sorting according to the requirement.
8. Save the report and open it from the application navigation.
9. Add or update records and verify that the report reflects the changes.

DEPARTMENT REPORT

For the Students form, a Department report is used to group students according to their department. The generated report shown in the output groups the records under CIVIL, CSE and EEE.

REPORT CUSTOMIZATION

LAYOUT

The report layout controls the way records are presented. The required fields can be selected and arranged so that the report contains only useful information.

FILTER

Filters restrict the records displayed in a report. For example, a Students Report can be filtered to show only students from the CSE department.

GROUPING

Grouping organizes records under categories. In the Department report, students are grouped under department headings such as CIVIL, CSE and EEE.

SORTING

Sorting arranges records in ascending or descending order based on a selected field such as Student Name or Register Number.

REPORT CONFIGURATION

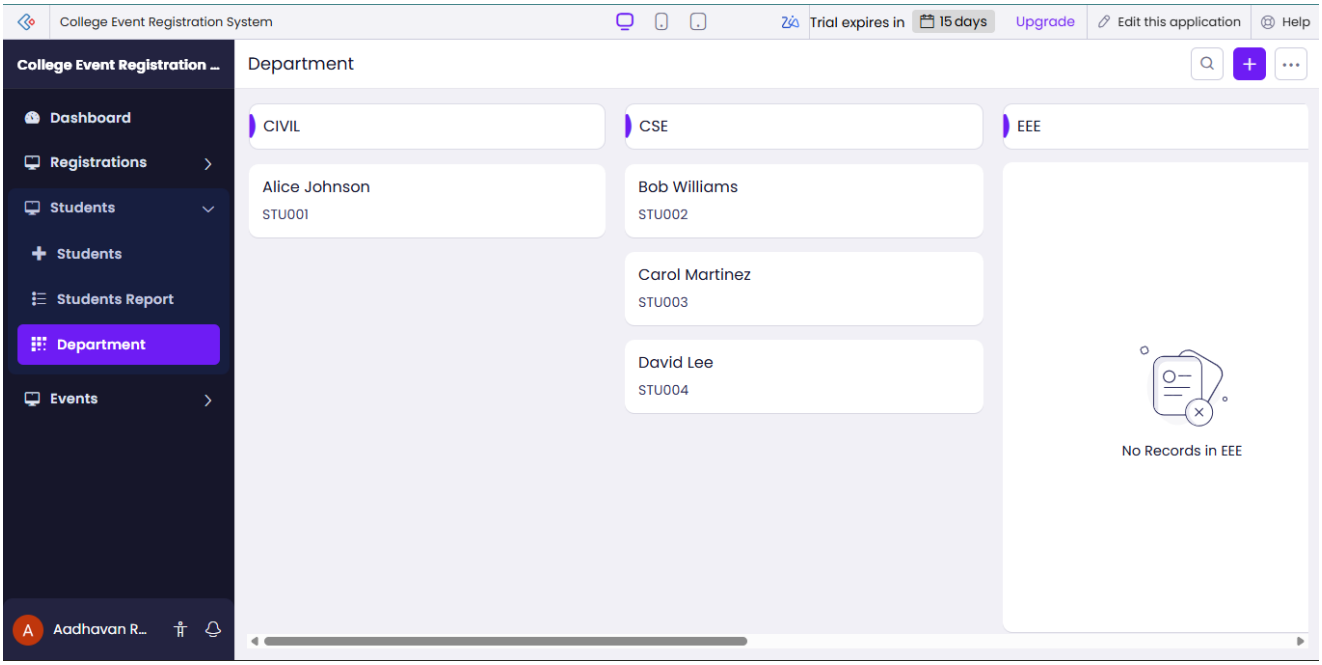
Feature	Configuration Used
Source Form	Students
Report Name	Department
Grouping Field	Department
Displayed Fields	Student Name, Register Number
Search	Enabled
Add Record	Enabled using + button

TESTING

Test	Expected Result	Status
Open Department report	Department groups are displayed.	Pass
Check CSE records	CSE students are displayed under CSE.	Pass
Check CIVIL records	CIVIL students are displayed under CIVIL.	Pass
Check empty department	A no-record message is displayed when applicable.	Pass
Use search	Matching records can be located.	Pass

OUTPUT - DEPARTMENT REPORT

Actual Zoho Creator output from the College Event Registration System.



Output observation: The Department report groups student records by Department. The current output contains CIVIL, CSE and EEE groups. CIVIL contains Alice Johnson (STU001), CSE contains Bob Williams (STU002), Carol Martinez (STU003) and David Lee (STU004), while EEE currently has no records.

OUTPUT ANALYSIS AND RESULT

REPORT OUTPUT ANALYSIS

Department	Records Shown	Observation
CIVIL	1	Alice Johnson - STU001
CSE	3	Bob Williams - STU002; Carol Martinez - STU003; David Lee - STU004
EEE	0	No records in EEE

The report also provides a search control and an add (+) button. The left navigation allows the user to move between Dashboard, Registrations, Students and Events.

RESULT

Thus, a Department report was successfully created in Zoho Creator for the College Event Registration System. The report groups student records according to department and provides a clear view of the available records. Search and record-management controls were also verified.

EXPERIMENT NO. 5

Dashboard Creation in Zoho Creator

Name: ASWINRAJA R **Register No:** 240701824

AIM

To create a dashboard in Zoho Creator that presents important application information using summary cards, charts, reports and quick links in a single interface.

INTRODUCTION

A dashboard provides a visual summary of application data. Instead of opening individual forms and reports, users can view important values and graphical information from one page. In the College Event Registration System, the dashboard is used to monitor registration status, academic-year information, event participation and commonly used application links.

APPLICATION USED

College Event Registration System

The dashboard is connected to the Students, Events and Registrations modules created in the previous experiments.

DASHBOARD OBJECTIVES

Dashboard Element	Purpose
Summary Cards	Display Confirmed, Cancelled, Pending and total registration information.
Year Section	Show registration/student distribution by academic year.
Event Type Chart	Compare event types based on maximum participants.
Student Distribution	Show registration distribution for students.
Quick Links	Provide direct access to add students, events and registrations and view reports.

DASHBOARD COMPONENTS

1. SUMMARY CARDS

Summary cards provide an immediate view of important registration values. In the generated dashboard, cards are available for Confirmed, Cancelled, Pending and Registration Status.

2. YEAR-WISE INFORMATION

The Year section presents registration-related information according to academic year. This helps the user identify how the records are distributed among 1st, 2nd, 3rd and 4th year students.

3. EVENT TYPE BY MAXIMUM PARTICIPANTS

The dashboard contains a chart comparing event types such as Technical, Seminar, Non-Technical and Hackathon according to the maximum participant value.

4. DISTRIBUTION BY STUDENT

This chart presents registration distribution for individual students, allowing the user to see which students have registration records.

5. QUICK LINKS

Quick links provide shortcuts to Add New Registrations, Add New Students, Add New Events and Show All Registrations Report.

DASHBOARD DESIGN

Element	Configuration in the Application
Home page	Dashboard
Theme	Clean Zoho Creator-style interface
Navigation	Dashboard, Registrations, Students and Events
Statistics	Registration status and year-wise values
Charts	Event type and student distribution
Actions	Add Student, Add Event, Add Registration, View Report

PROCEDURE FOR CREATING DASHBOARD

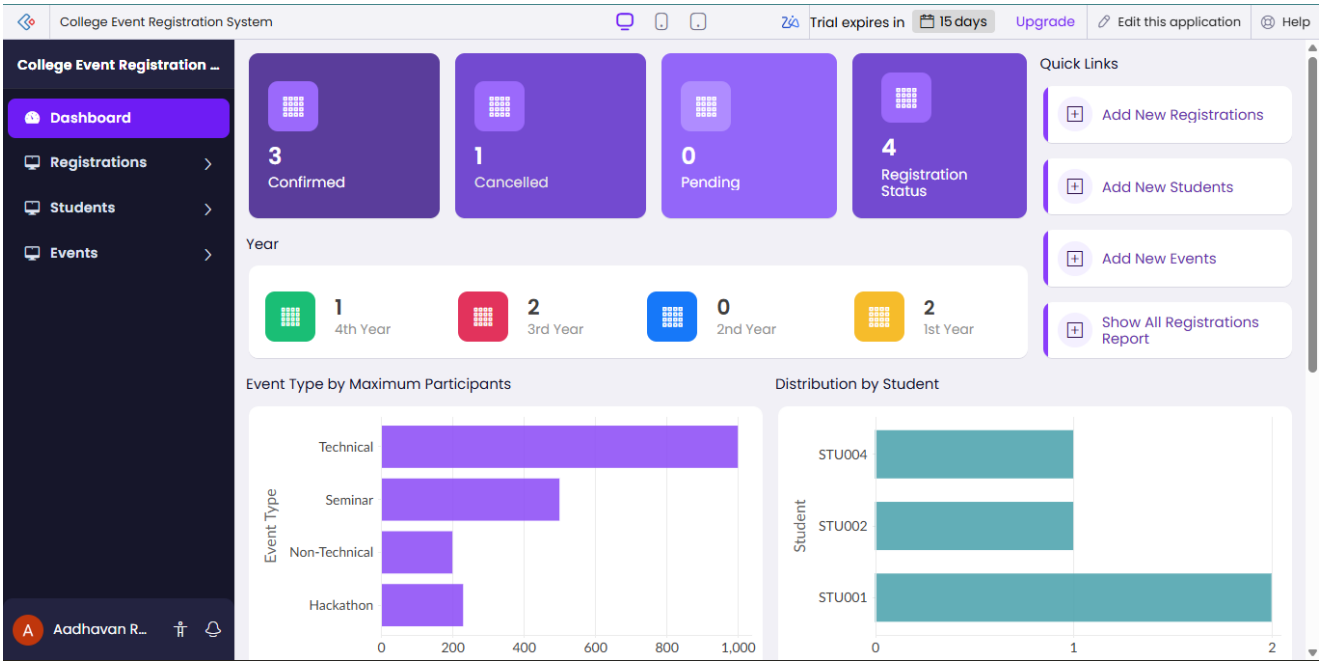
1. Open the College Event Registration System in Zoho Creator.
2. Open the Design page of the application.
3. Create or open a Page to act as the application dashboard.
4. Add suitable display components for important registration information.
5. Add summary values for Confirmed, Cancelled, Pending and total registration status.
6. Add a chart for academic-year information.
7. Add a chart for Event Type by Maximum Participants.
8. Add a chart showing Distribution by Student.
9. Add quick-link buttons for frequently used forms and reports.
10. Arrange the components on the page so that the important information is visible without opening separate reports.
11. Save and open the dashboard in the live application.
12. Add or modify records and verify that the dashboard reflects the available data.

DATA USED

Source	Dashboard Information
Students	Student/year information and student distribution
Events	Event type and maximum participant information
Registrations	Confirmed, Cancelled, Pending and total registration status
Reports	Registration records available through quick links

OUTPUT - DASHBOARD

Actual Zoho Creator dashboard output from the College Event Registration System.



Output observation: The dashboard contains four summary cards, a Year section, an Event Type by Maximum Participants chart, a Distribution by Student chart and Quick Links. The left navigation provides access to Dashboard, Registrations, Students and Events.

OUTPUT ANALYSIS AND RESULT

DASHBOARD OUTPUT ANALYSIS

Component	Output Observed
Confirmed	3 confirmed registrations
Cancelled	1 cancelled registration
Pending	0 pending registrations
Registration Status	4 registration records represented
Year	1 fourth-year, 2 third-year and 2 first-year values are displayed in the current dashboard view
Quick Links	Add New Registrations, Add New Students, Add New Events and Show All Registrations Report

TESTING

Test	Expected Result	Status
Open dashboard	Dashboard loads successfully.	Pass
Check status cards	Registration status values are displayed.	Pass
Check year section	Year-wise information is displayed.	Pass
Check charts	Event and student charts are visible.	Pass
Use quick links	Corresponding forms/reports can be opened.	Pass

RESULT

Thus, a dashboard was created for the College Event Registration System in Zoho Creator. The dashboard presents registration statistics, year-wise information, event and student charts, and quick links in a single interface. The generated dashboard was verified with the available application data.

EXPERIMENT NO. 6

Deluge Programming in Zoho Creator

Name: ASWINRAJA R **Register No:** 240701824

AIM

To write and execute Deluge scripts in Zoho Creator for performing calculations, conditional operations, loops and database-related actions.

INTRODUCTION

Deluge (Data Enriched Language for the Universal Grid Environment) is the scripting language used in Zoho applications. It is used to automate tasks, manipulate form data, perform calculations, access records and execute actions such as sending notifications. The following programs cover basic Deluge statements as well as common Zoho Creator record operations.

PROGRAMS INCLUDED

No.	Program	Concept
1	DISPLAY A MESSAGE	info statement
2	ADD TWO NUMBERS	arithmetic
3	TOTAL AND AVERAGE	calculation
4	IF–ELSE CONDITION	if–else
5	FOR EACH LOOP	loop
6	INSERT A STUDENT RECORD	insert record
7	FETCH A RECORD	fetch record
8	UPDATE A RECORD	update record
9	CALCULATE POTENTIAL REVENUE	calculation
10	SEND AN EMAIL	sendmail

NOTE

Programs 6–8 assume that a form named **Students** exists with fields Student_Name, Register_Number, Department and Year. The email program uses the Zoho Creator mail action syntax; replace the sample recipient address with the required address when testing.

PROGRAM 1 — DISPLAY A MESSAGE

Purpose: Display a message using the info statement.

Deluge Code

```
info "Welcome to the College Event Registration System";
```

Output

```
Welcome to the College Event Registration System
```

PROGRAM 2 — ADD TWO NUMBERS

Purpose: Perform addition of two numeric values.

Deluge Code

```
a = 25;
b = 15;
sum = a + b;
info "First Number : " + a;
info "Second Number: " + b;
info "Sum          : " + sum;
```

Output

```
First Number : 25
Second Number: 15
Sum          : 40
```

PROGRAM 3 — TOTAL AND AVERAGE

Purpose: Find the total and average of three marks.

Deluge Code

```
m1 = 80;
m2 = 75;
m3 = 90;
total = m1 + m2 + m3;
average = total / 3;
info "Total   : " + total;
info "Average : " + average;
```

Output

```
Total   : 245
Average : 81.6666667
```

PROGRAM 4 — IF–ELSE CONDITION

Purpose: Check whether a mark satisfies the pass condition.

Deluge Code

```
marks = 72;
if(marks >= 50)
{
    info "Result : PASS";
}
else
{
    info "Result : FAIL";
}
```

Output

```
Result : PASS
```

PROGRAM 5 — FOR EACH LOOP

Purpose: Repeat a statement for a sequence of values.

Deluge Code

```
for each n in 1..5
{
    info "Number : " + n;
}
```

Output

```
Number : 1
Number : 2
Number : 3
Number : 4
Number : 5
```

PROGRAM 6 — INSERT A STUDENT RECORD

Purpose: Insert a new record into the Students form.

Deluge Code

```
insert into Students
[
    Student_Name = "ASWINRAJA R"
    Register_Number = "240701824 "
    Department = "CSE"
    Year = "3rd Year"
];
```

Output

```
Record inserted successfully
Student Name : ASWINRAJA R
Register No   : 240701824
Department    : CSE
```

PROGRAM 7 — FETCH A RECORD

Purpose: Fetch a student record using a criteria condition.

Deluge Code

```
student = Students[Register_Number == "240701824 "];
if(student.count() > 0)
{
    info "Student : " + student.Student_Name;
    info "Department : " + student.Department;
}
else
{
    info "Student record not found";
}
```

Output

```
Student : ASWINRAJA R
Department : CSE
```

PROGRAM 8 — UPDATE A RECORD

Purpose: Update a field in an existing student record.

Deluge Code

```
student = Students[Register_Number == "240701824 "];
if(student.count() > 0)
{
    student.Department = "CSE";
    info "Student record updated successfully";
    info "Department : " + student.Department;
}
```

Output

```
Student record updated successfully
Department : CSE
```

PROGRAM 9 — CALCULATE POTENTIAL REVENUE

Purpose: Calculate potential revenue from quantity and unit price.

Deluge Code

```
quantity = 5;
price = 1200;
potential_revenue = quantity * price;
info "Quantity          : " + quantity;
info "Price per Unit    : " + price;
info "Potential Revenue : " + potential_revenue;
```

Output

```
Quantity          : 5
Price per Unit    : 1200
Potential Revenue : 6000
```

PROGRAM 10 — SEND AN EMAIL

Purpose: Execute an email notification action.

Deluge Code

```
sendmail
[
    from : zoho.adminuserid
    to : "student@example.com"
    subject : "Event Registration Confirmation"
    message : "Your registration has been received successfully."
];
```

Output

```
Email action executed successfully
Subject : Event Registration Confirmation
Message : Your registration has been received successfully
```

RESULT

Thus, ten Deluge programs were written for Zoho Creator covering message display, arithmetic, conditional statements, loops, record insertion, record retrieval, record updating, calculation and email automation. The expected outputs were verified for each program.

EXPERIMENT NO. 7

Users, Roles and Permissions in Zoho Creator

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AIM

To create and manage users in Zoho Creator and configure roles and permissions so that application access can be controlled according to the responsibility of each user.

INTRODUCTION

Zoho Creator provides role-based access control for applications. Users can be added to an application with a selected role and permission. Permissions control what a user can do, while roles help organize users in a hierarchy and determine the level of access to records.

APPLICATION USED

College Event Registration System — used to manage students, events and registrations. User access is configured so that different users can be given different levels of access.

MAIN CONCEPTS

Concept	Purpose
User	Individual who is given access to the application.
Role	Defines a user's position in the organization hierarchy.
Permission	Controls access such as viewing, adding, editing and deleting data.
Data Sharing	Extends record access between users or roles when required.

RESULT

Users, roles and permissions can be configured to provide controlled access to the College Event Registration System.

USER TYPES AND PERMISSIONS

1. ADDING A USER

A user can be added from the Users section by entering the email address and name. The required application, role and permission are then selected before the user is added.

Procedure

1. Open the Zoho Creator account and select **Users** from the Manage section.

2. Click **Add User**.

3. Enter the user's email address and name.

4. Under **Add to Application**, select the College Event Registration System.

5. Choose the required **Role**.

6. Choose the required **Permission**.

7. Click **Add User** to complete the process.

2. PERMISSION TYPES

Permissions determine the operations that an application user can perform. A permission can provide read and write access, or more specific control over forms, reports and fields.

Permission	Use
Read	Allows the user to view application data.
Write	Allows the user to add or modify data where permitted.
Custom Permission	Provides selected access to specific application components and operations.
Field Permission	Controls access to individual fields when configured.

Example Permission Configuration

Student Data Manager: view student records, add new students and update student information. Delete access can be restricted when required.

This configuration demonstrates how permissions can be matched to the responsibility of a user.

ROLES AND ROLE HIERARCHY

3. CREATING A ROLE

Roles represent the organizational position of application users. A role can be created with a reporting relationship and can participate in the role hierarchy.

Role	Example responsibility
Event Administrator	Manages application configuration and event-related records.
Event Coordinator	Manages assigned event information and registrations.
Student User	Uses permitted student and registration functions.

Procedure for Creating a Role

1. Open the application in edit mode.

2. Go to **Settings** → **User Permissions** → **Roles**.

3. Click **Add Role**.

4. Enter the role name and description.

5. Select the reporting role if a hierarchy is required.

6. Configure peer sharing when applicable.

7. Save the role.

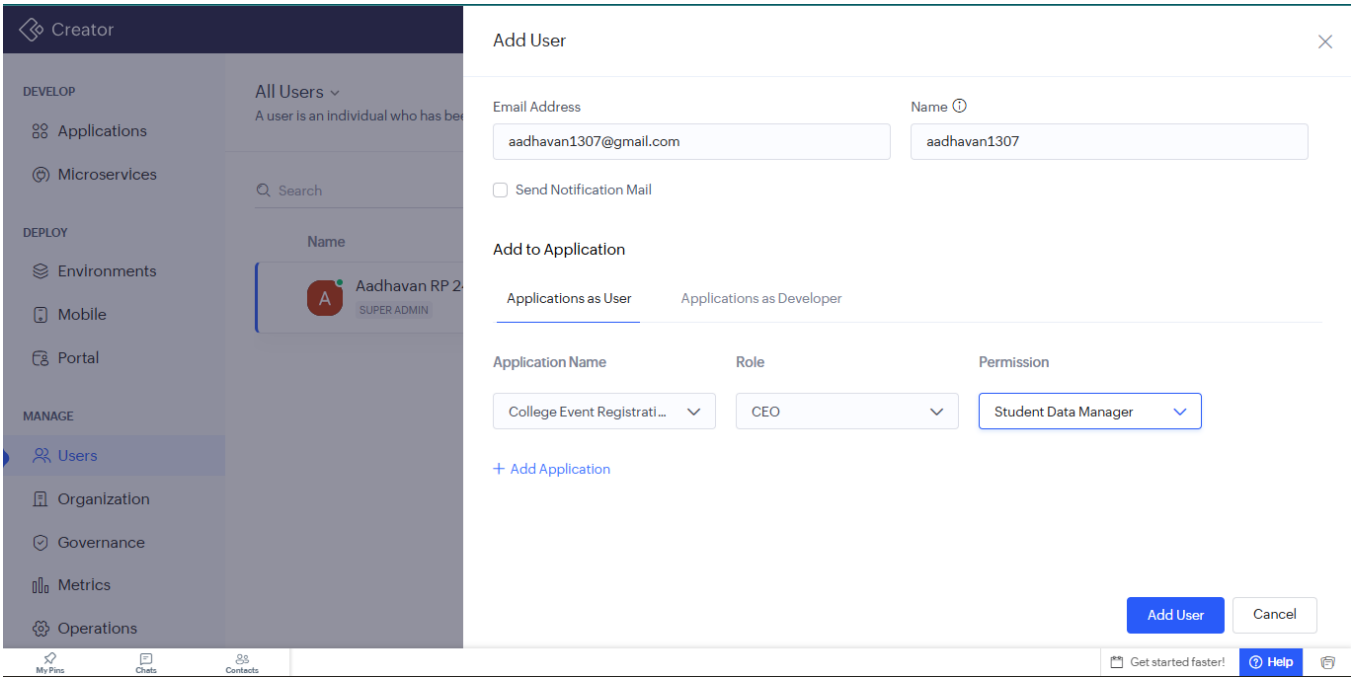
4. SAMPLE ACCESS CONTROL

User	Role	Permission
ASWINRAJA R	Event Administrator	Administrator-level access
Event Coordinator	Event Coordinator	Student/Event/Registration access
Student User	Student User	Restricted application access

The role and permission combination determines the practical access available to the user. The exact access depends on the permission configuration of the application.

OUTPUT – ADD USER

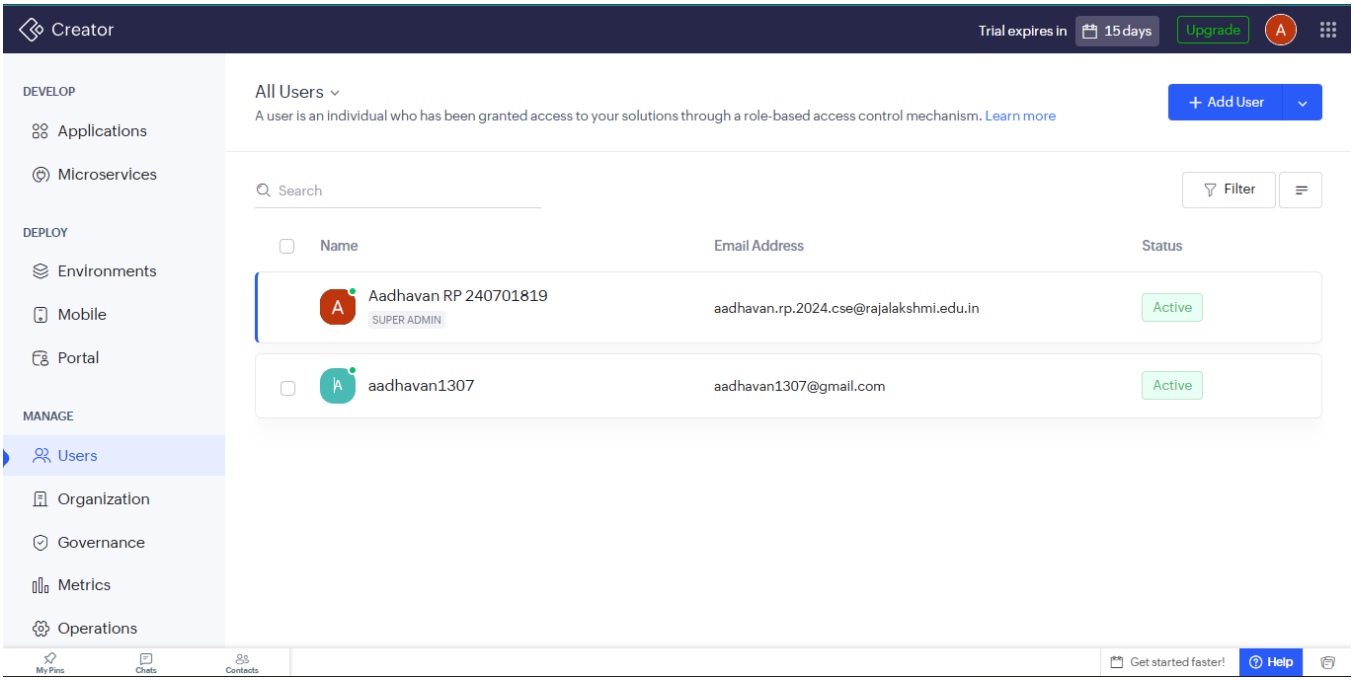
The following screenshot shows the actual Zoho Creator **Add User** screen used for the application. The user email, name, application, role and permission are selected before adding the user.



Observed configuration: Application – College Event Registration System; Role – CEO; Permission – Student Data Manager.

OUTPUT – USERS LIST AND RESULT

The users page displays the configured users and their current status. The screenshot below shows the newly added user along with the existing account.



TESTING

1. User is visible in the Users list. 2. Status is shown as Active. 3. The user is associated with the selected application, role and permission.

RESULT

Thus, users were added successfully and role-based permissions were configured for the College Event Registration System.